Warmup/Attendance: [Week 7: Wednesday 10/7/2026]

Full Name: | SID:

Work with the people around you to try to solve these problems, which we'll discuss together once lecture begins. These are not graded for correctness, but you will turn this sheet in for attendance at the end of class.

1. Comparing expected values

T is a random variable that takes the value 3 with probability $1/3$, the value 4 with probability $1/4$, and the value 5 with the remaining probability.

Now consider the box with tickets: 3 3 3 4 4 4 4 5 5 5 5 5

Suppose we draw **once** from this box and let U be the value of the ticket drawn. Fill in the blank with "less than", "greater than", or "equal to":

The expected value of T is ____ the expected value of U.

2. Office hours and the Poisson distribution

A prof notices that their office hours are not too crowded this semester. They observe that a random variable X representing the number of *Stat 20* students coming to their weekly office hours has a Poisson(2) distribution. Furthermore, there is one Data 88 student from a previous semester who is *always* there (they want a letter of recommendation).

Let V be the *total* number of students in their office hours. What are the expected value and variance of V?

3. What does squaring do to the expectation and variance?

Let X be a discrete uniform random variable on the set $\{-1, 0, 1\}$.

If $Y = X^2$, what are $E(Y)$ and $Var(Y)$?

4. Minimum function

The *minimum* function $\min(a, b)$ takes in two numbers a and b and outputs the *smaller* of the two. Let X be a discrete uniform random variable on the set $\{-1, 0, 1\}$.

If $W = \min(X, 0.5)$, what is $E(W)$?